Experiment No: 1 (Software) **Date: 21-08-2022**

Verification of Logic Gates using Verilog

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Aim: To Verify the truth table of the logic gates in Verilog using Gate, dataflow and Behavioural Modelling.

Software required: ModelSim

Procedure:

1. Click Jumpstart, create new project and type the project name and click ok.
2. Click Create New File, type the name and click Verilog from the drop-down and click ok.
3. After typing the code, click compile, compile selected from menu.
4. Click Simulate and click start simulation.
5. For input, select the input using Force or Clock and run it.

Verilog Code:

Gate Modelling

```
module gates (a,b,c,d,e,f,g,h,i);

input a,b;

output c,d,e,f,g,h,i;

and o1(c,a,b);

or o2(d,a,b);

not o3(e,a);

xor o4(f,a,b);

xnor o5(g,a,b);

nand o6(h,a,b);

nor o7(i,a,b);
```

```
endmodule
```

Data flow Modelling

```
module gates (a,b,c,d,e,f,g,h,i);
input a,b;
output c,d,e,f,g,h,I;
assign c = a&b;
assign d = a|b;
assign e = ~a;
assign f = a^b;
assign g = a~^b;
assign h = ~(a&b);
assign I = ~(a|b);
endmodule
```

Behavioral Modelling

<u>AND Gate:</u>

```
module gates(a,b,c);
input a,b;
output reg c;
always @ (a or b) begin
  if (a==1'b1 & b==1'b1) begin
    c=1'b1;
  end
  else
    c=1'b0;
end
```

```
endmodule
```

OR Gate:

```
module gates(a,b,c);
input a,b;
output reg c;
always @ (a or b) begin
  if (a==1'b1 | b==1'b1) begin
    c=1'b1;
  end
  else
    c=1'b0;
end
endmodule
```

NOT Gate:

```
module gates(a,c);
input a;
output reg c;
always @ (a) begin
  if (a == 1'b1) begin
    c=1'b0;
  end
  else
    c=1'b1;
end
endmodule
```

XOR Gate:

```
module gates(a,b,c);
input a,b;
output reg c;
always @ (a or b) begin
  if (a==b) begin
    c=1'b0;
  end
  else
    c=1'b1;
end
endmodule
```

XNOR Gate:

```
module gates(a,b,c);
input a,b;
output reg c;
always @ (a or b) begin
  if (a==b) begin
    c=1'b1;
  end
  else
    c=1'b1;
end
endmodule
```

NAND Gate:

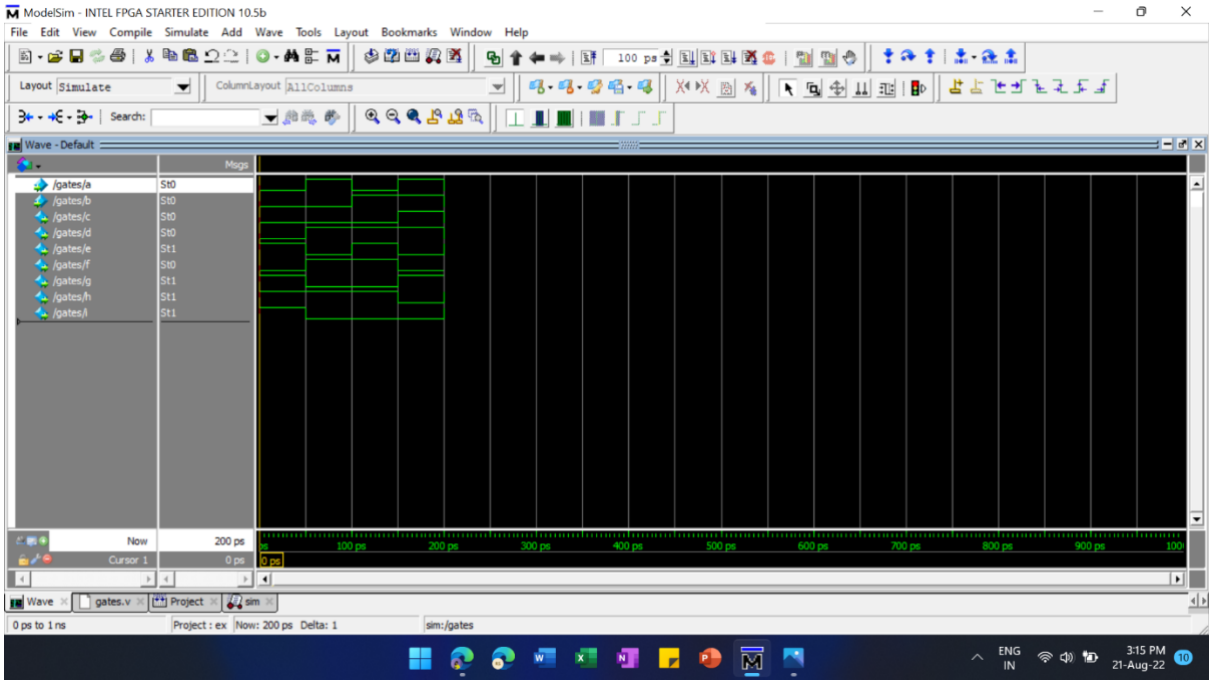
```
module gates(a,b,c);
input a,b;
output reg c;
always @ (a or b) begin
  if (a==1'b1 & b==1'b1) begin
    c=1'b0;
  end
  else
    c=1'b1;
end
endmodule
```

NOR Gate:

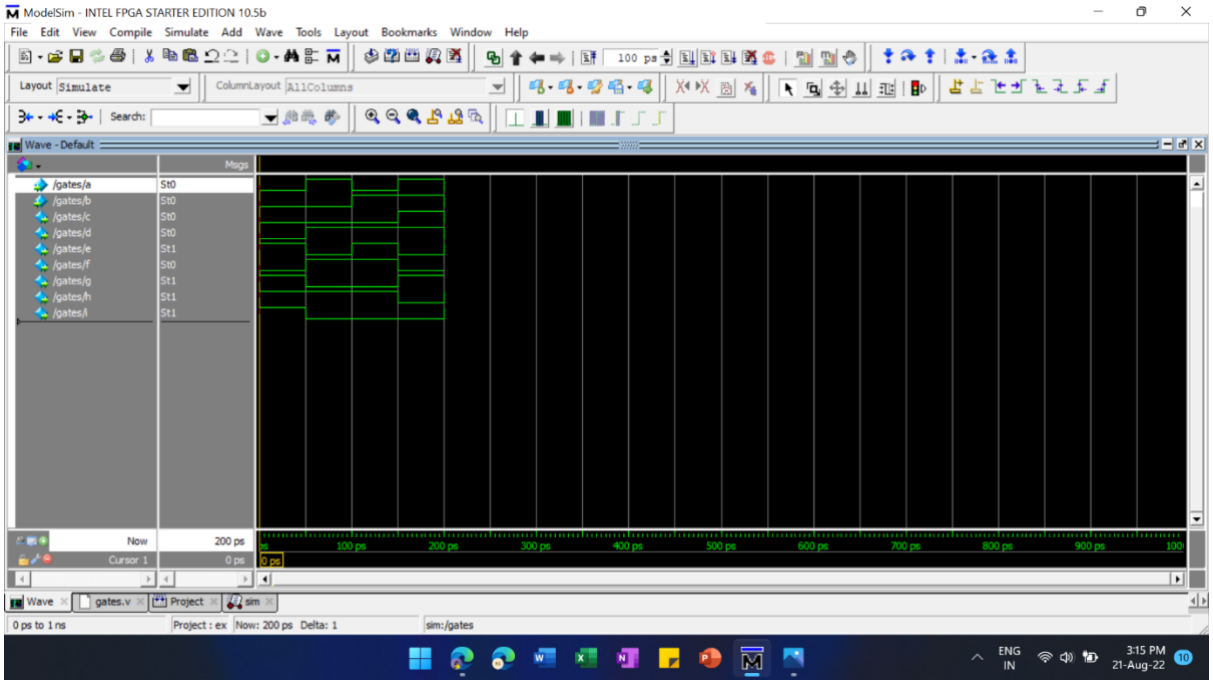
```
module gates(a,b,c);
input a,b;
output reg c;
always @ (a or b) begin
  if (a==1'b1 & b==1'b1) begin
    c=1'b0;
  end
  else
    c=1'b1;
end
endmodule
```

Verilog Output:

Gate Modelling

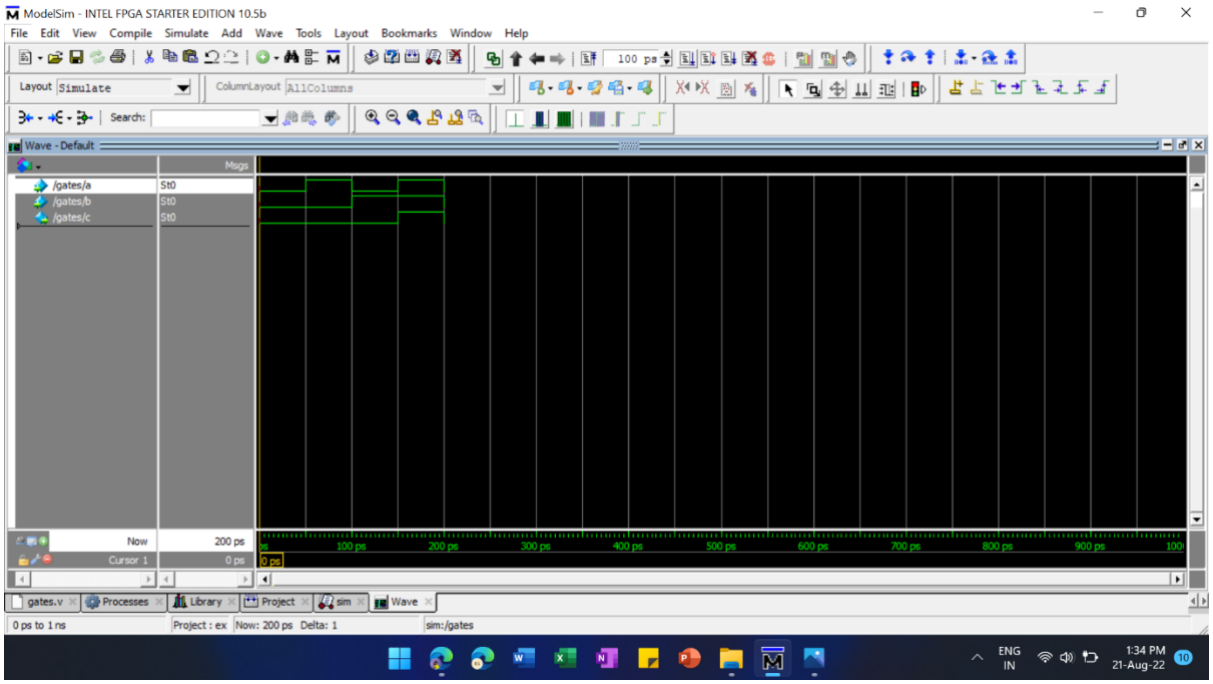


Data flow Modelling

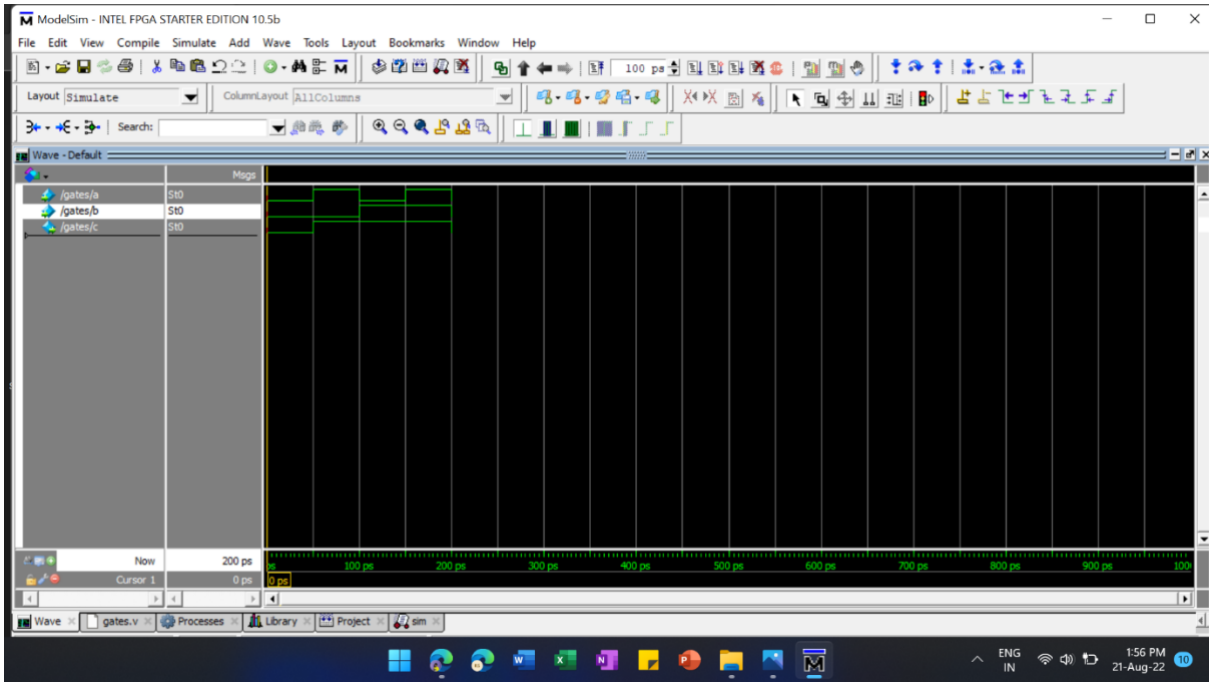


Behavioral Modelling

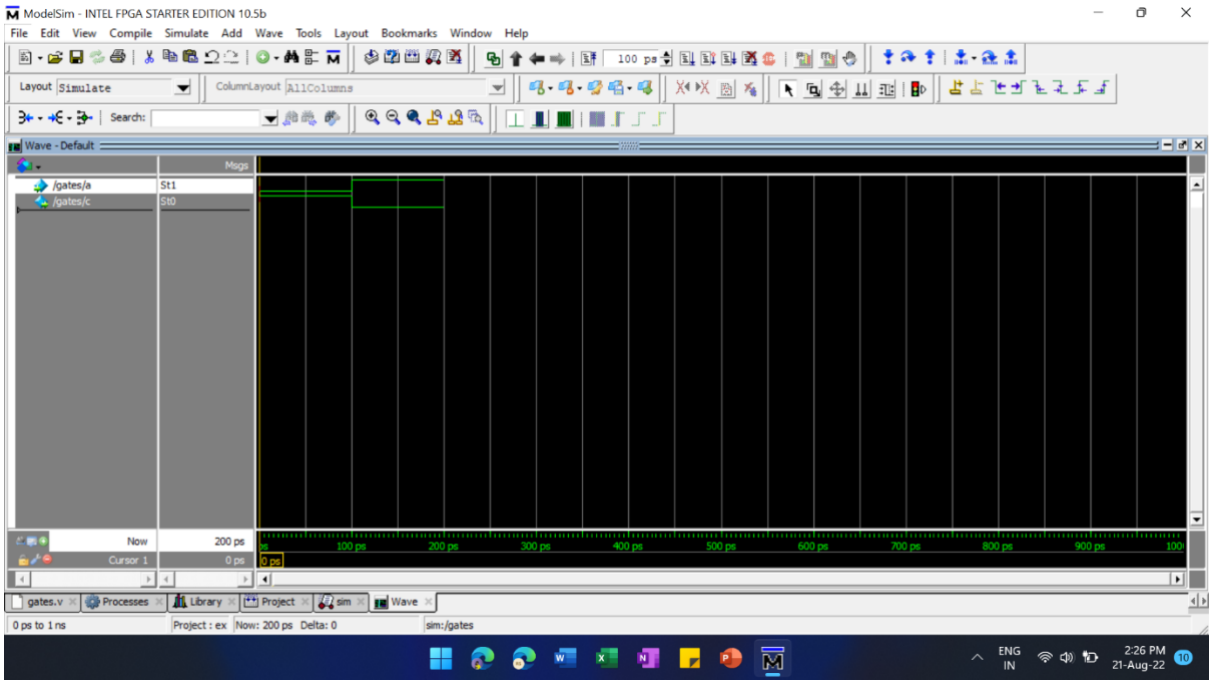
AND Gate:



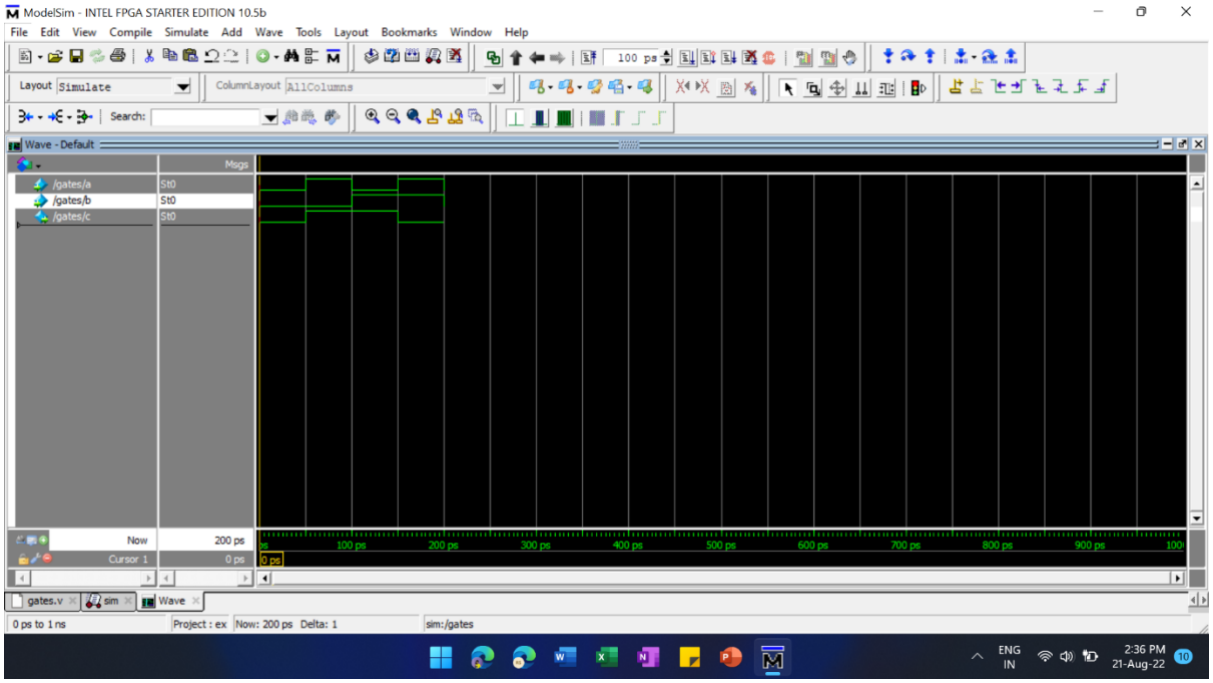
OR Gate:



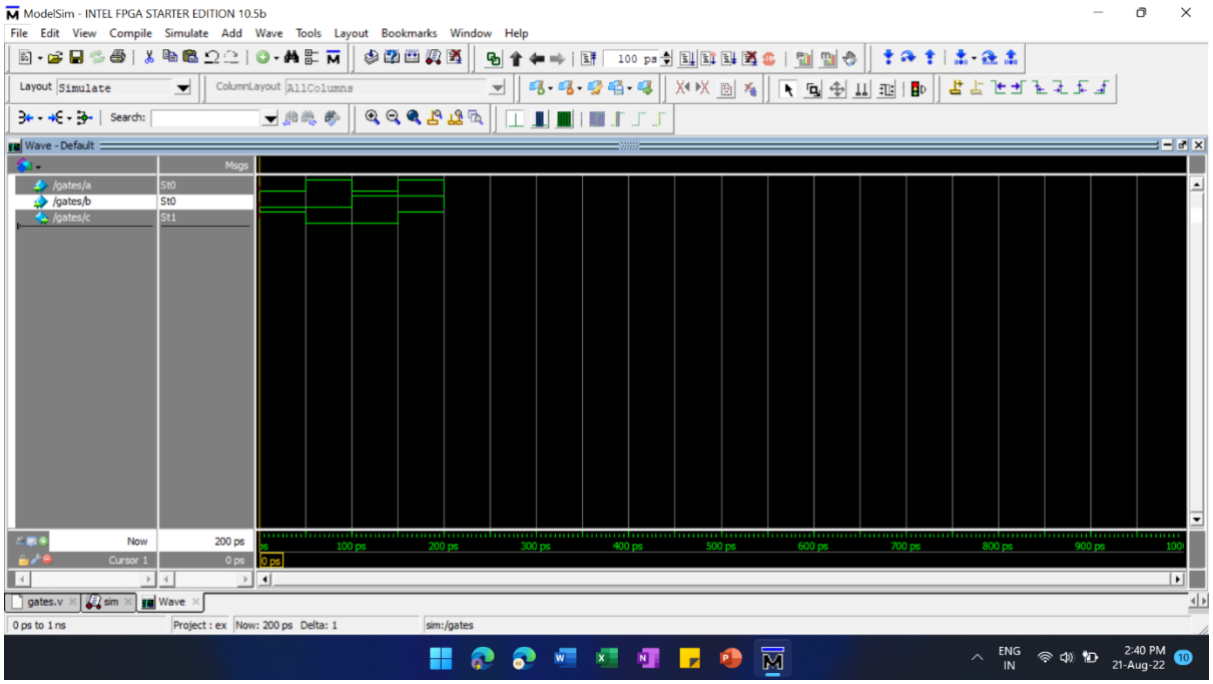
NOT Gate:



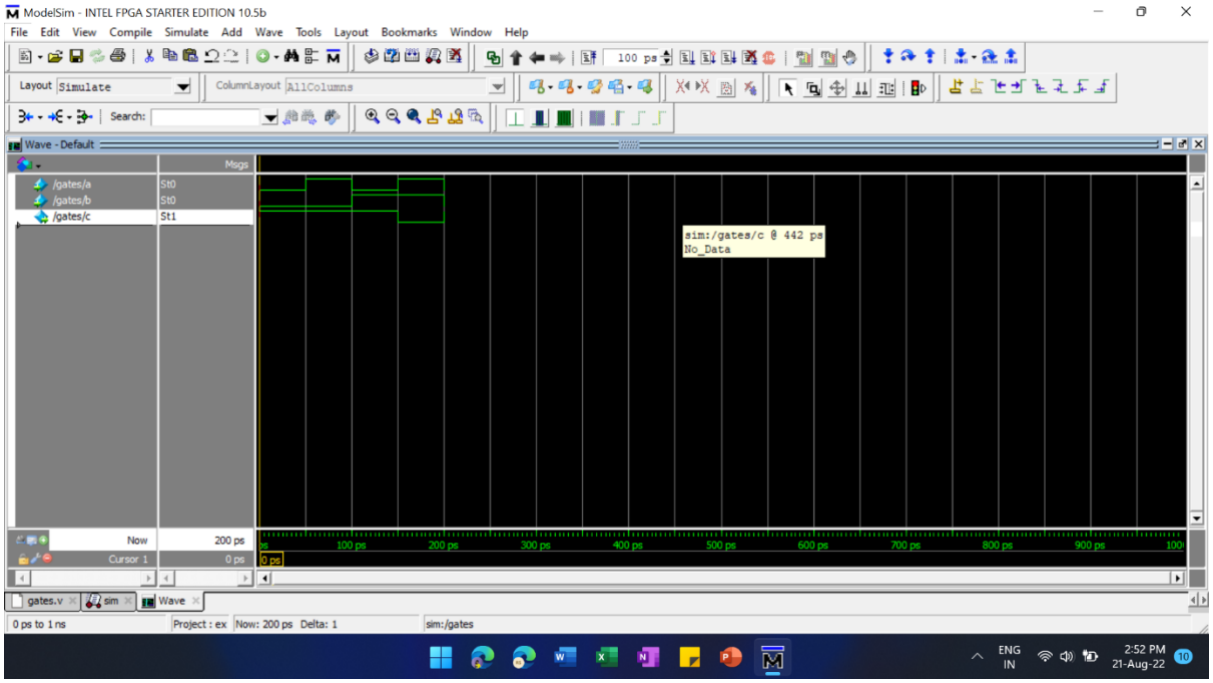
XOR Gate:



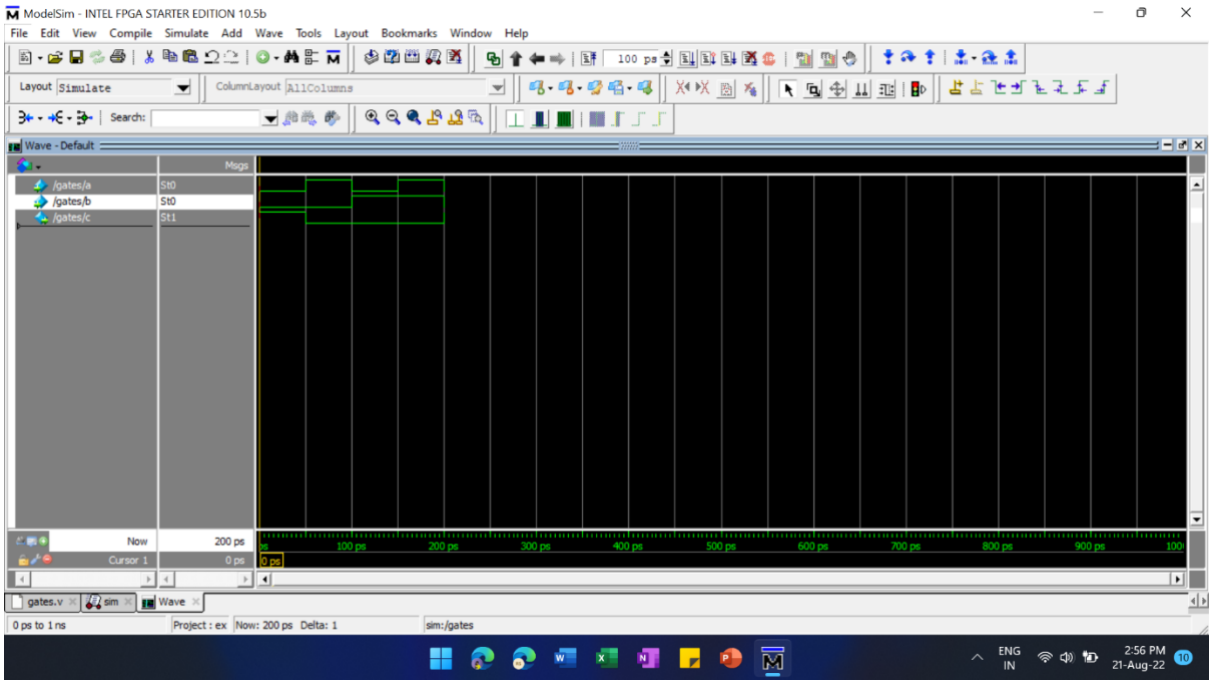
XNOR Gate:



NAND Gate:



NOR Gate:



Result/Inference:

From this experiment, the truth table of the logic gates in Verilog using Gate, dataflow and Behavioural Modelling is verified.